

Blood Bank Unit Allocation System

Product Requirements Document | Program Design & Development

Problem Statement & Description

A regional blood bank receives requests from hospitals for blood units of eight different blood groups, but has only a limited stock of each group available, and every unit also has a shelf life of about five weeks before it expires. This command-line app lets the bank's staff plan the day's distribution — adding new stock, logging hospital requests, and letting the system decide which requests to fulfil, with which exact units, and at what fee — and keeps track of which units are issued, reserved, or wasted.

Interface

Inputs:

- Menu choice: a number from 1 to 7 selecting the action to perform.
- Add stock: one or more blood group and expiry date pairs, comma-separated (e.g., O+ 2026-10-01, A- 2026-09-20). The expiry date must always be entered — it is not optional.
- Submit request: hospital name, hospital type (Private/Govt), blood group (one exact group, e.g. O- or O+), quantity, and urgency (High/Mid/Low).
- Dates are entered in YYYY-MM-DD format (e.g., 2026-10-01). A date that doesn't match this format is treated as invalid — the unit is not added.
- A valid expiry date must fall within 35 days of today (inclusive). This reflects the typical shelf life of stored whole blood and red cell units, so the app won't accept a date implying the unit would still be usable well beyond that window.
- Hospital type, blood group, and urgency are accepted case-insensitively (e.g. "govt" and "Govt" are treated the same), but must otherwise match one of the fixed valid values exactly.
- Hospital name cannot be left blank; an empty entry is rejected with an error.

Outputs:

- The stock table: all eight blood groups — A+, A-, B+, B-, AB+, AB-, O+, O- — along with how many units of each are currently available and the nearest expiry date among them.
- The menu: seven options — 1. Add stock, 2. Submit request, 3. Generate allocation plan, 4. Confirm & issue plan, 5. View progress report, 6. View expired/wasted report, 7. Exit.
- The stock table and menu are printed again after each action, until staff choose Exit.

Functional Requirements

Here are the nine functional requirements that must be satisfied by the command-line app. An example interaction is provided along with each functional requirement.

FR 1: Print the stock table, then the menu.

Example:

```
Group Available Nearest Expiry
A+ 5 2026-09-20
A- 2 2026-09-15
B+ 3 2026-09-25
B- 0 -
AB+ 1 2026-10-01
```

AB- 0 -
O+ 6 2026-09-18
O- 4 2026-09-12
Pending requests: 0

1. Add stock
2. Submit request
3. Generate allocation plan
4. Confirm & issue plan
5. View progress report
6. View expired/wasted report
7. Exit
Choose an option:

FR 2: When staff choose Add stock, let them add one or more units by blood group and expiry date.

Rule:

- Each accepted unit is given a unique ID and added to that group's available stock.
- If an entered expiry date is already in the past, that unit is not added (see the corresponding test case).
- If an entered expiry date is more than 35 days from today, that unit is also not added, rejected the same way as a past date. 35 days is used as the cutoff because it matches the shelf life generally cited for stored whole blood and red cell units, so the app never accepts a date implying blood that's good for longer than that.
- A date exactly 35 days from today is valid and accepted.

Example:

Choose an option: 1
Enter group and expiry date (e.g., O+ 2026-10-01), comma-separated: O- 2026-09-20, A+ 2026-11-05
Added: U011 O- (exp 2026-09-20)
Error: Expiry date 2026-11-05 is more than 35 days from today. Unit not added.

FR 3: When staff choose Submit request and the urgency is High, check the request immediately; if enough compatible units are available, issue them and print an itemized bill.

Rule:

- A unit can only be issued to a matching blood group. Compatibility — O- can go to any group; O+ can go to O+, A+, B+, AB+; A- can go to A-, A+, AB-, AB+; A+ can go to A+, AB+; B- can go to B-, B+, AB-, AB+; B+ can go to B+, AB+; AB- can go to AB-, AB+; AB+ can go to AB+ only.
- All of the requested quantity must be available from compatible stock. If it is, issue the compatible units closest to expiry first, mark the request Fulfilled, and print a bill: each issued unit with its fee, the transportation charge, then the total.
- If not enough compatible units exist, mark the request Rejected — print how many compatible units are actually available, but issue nothing and charge no fee.
- Fee components applied per issued unit: Base Fee: Rs 500 on every issued unit, always. Urgency Adjustment (per unit): High +Rs 300, Mid +Rs 100, Low +Rs 0. Blood Group Adjustment (per unit): Rh-negative (A-, B-, AB-, O-) +Rs 150, Rh-positive (A+, B+, AB+, O+) +Rs 0. Hospital Type Discount (per unit): Private -Rs 0, Govt -Rs 150. Transportation Charge: Rs 100 flat, added once per fulfilled request only (not per unit).
- Per-unit fee = 500 + Urgency Adjustment + Blood Group Adjustment - Hospital Discount. Total bill = (sum of per-unit fees) + Rs 100 transportation.
- Mid and Low urgency requests are not checked immediately — they are queued and handled together under Generate allocation plan (FR 4).

- When more than one blood group is compatible with a request, units are picked strictly by soonest expiry across all compatible groups — not by preferring an exact group match over a cross-compatible one (e.g. O-).
- If two compatible units share the same nearest expiry date, the one with the earlier unit ID is issued first.

Example:

```
Choose an option: 2
Hospital name: City Hospital
Hospital type (Private/Govt): Private
Blood group: O-
Quantity: 5
Urgency (High/Mid/Low): High
```

----- BILL -----

Unit ID	Group	Expiry	Fee
U004	O-	2026-09-12	Rs 950
U011	O-	2026-09-20	Rs 950
U015	O-	2026-09-22	Rs 950
U019	O-	2026-09-25	Rs 950
U020	O-	2026-09-28	Rs 950

Transportation Charge: Rs 100

TOTAL AMOUNT: Rs 4850

Status: Fulfilled

```
Choose an option: 2
Hospital name: City Hospital
Hospital type (Private/Govt): Private
Blood group: O-
Quantity: 30
Urgency (High/Mid/Low): High
Requested: 30 units of O-
Available (compatible): 5 units
Status: Rejected - insufficient stock (need 30, have 5)
No units issued. No fee charged.
```

FR 4: When staff choose Generate allocation plan, process all pending Mid and Low requests together and propose an allocation for each.

Rule:

- Process Mid requests before Low requests, and within each urgency, process them in the order they were received.
- For each request, apply the same compatibility and all-or-nothing rule as FR 3: if enough compatible units are available, propose the closest-to-expiry units and mark it Fulfilled with a bill; otherwise mark it Rejected and show how many compatible units are available.
- Nothing is issued yet — this only prints a proposed plan and a proposed total revenue.
- Requests are processed one at a time in the batch order (Mid before Low, received order within each). Once a request consumes units, those units are no longer available to the next request in the same batch — so a later request needing the same group may be Rejected even though it would have been fulfillable on its own.

Example:

```
Choose an option: 3
Request R1 (District Hospital, Govt, A+, Qty 2, Low)
Proposed: U007 A+ (exp 2026-09-18) Rs 350
U012 A+ (exp 2026-10-05) Rs 350
Transportation: Rs 100
Status: Fulfilled - Total Rs 800

Request R2 (Metro Clinic, Private, AB-, Qty 2, Mid)
Status: Rejected - insufficient stock (need 2, have 0)

Proposed Total Revenue: Rs 800
Choose option 4 to confirm and issue this plan.
```

FR 5: When staff choose Confirm & issue plan, commit the most recently generated plan.

Rule:

- Every unit proposed in the plan is marked Issued and removed from available stock.
- Each request's status becomes final, its fee is added to the session's total revenue, and it is removed from the pending queue.

Example:

```
Choose an option: 4
Plan confirmed. Units issued, stock updated.
Total revenue this session: Rs 2800
```

FR 6: Automatically expire unissued units once their expiry date has passed.

Rule:

- Before the stock table is printed each time, any available unit whose expiry date has passed is removed from stock and logged as wasted.
- Each wasted unit's base fee (Rs 500) is logged as a loss, shown in the expired/wasted report (FR 8).

Example:

```
[2 units expired and discarded: U002 (A-, exp 2026-09-05), U008 (O+, exp 2026-09-06)]
```

FR 7: When staff choose View progress report, print how the day is going so far.

Rule:

- Shows total requests handled, broken down into Fulfilled and Rejected counts, and total revenue earned so far in the session.
- This report does not change any state — it only reads and displays current totals.

Example:

```
Choose an option: 5
Requests handled: 5 total
Fulfilled: 3
Rejected: 2
Total Revenue so far: Rs 3600
```

FR 8: When staff choose View expired/wasted report, print every unit lost to expiry and its total value.

Rule:

- Lists each wasted unit with its ID, blood group, and the date it expired.
- Shows the total number of wasted units and the total value lost (Rs 500 base fee per wasted unit).

Example:

```
Choose an option: 6
Expired / Wasted Units: 2
U002 A- expired 2026-09-05
U008 O+ expired 2026-09-06
Total Value Lost: Rs 1000
```

FR 9: After each action, return to the stock table and menu; when staff choose Exit, stop.

Rule:

- Units booked or expired earlier stay that way for every action after. After an action (or an error), print the updated stock table and menu again.
- When staff choose Exit, the app stops.

Example:

```
Choose an option: 7
Goodbye
```

Edge Cases

#	Input / Situation	Expected Behaviour
EC1	Invalid blood group (e.g., C+, AB) on Submit Request	Print an error naming the invalid group; request rejected.
EC2	Quantity is zero, negative, or non-numeric	Print an error; request rejected.
EC3	Urgency is not High/Mid/Low	Print an error; request rejected.
EC4	Hospital type is not Private/Govt	Print an error; request rejected.
EC5	No compatible stock exists at all for a request	Print rejection naming zero available units; status = Rejected.
EC6	Stock added with an expiry date already in the past	Print an error; unit not added. Same rejection path as a too-far-future date (EC16) — both are “outside the valid window.”
EC7	“Confirm & issue” chosen with no plan generated	Print an error; no state change.
EC8	“Generate plan” chosen with zero pending requests	Print a message; nothing generated.
EC9	Menu choice not in 1-7 (e.g., 8, blank)	Print an error and show the menu again.
EC10	Invalid blood group entered on Add Stock (e.g., C+, AB)	Print an error naming the invalid group; that unit is not added, remaining valid pairs still processed.
EC11	Compatible stock exists but is less than the requested quantity (partial stock)	Print rejection showing available vs. needed count; status = Rejected; no partial issue of the available units.
EC12	A unit's expiry date is exactly today's date	Treat the unit as still valid (not expired); it stays Available and can still be issued.
EC13	Stock changes (e.g., a proposed unit expires) between Generate plan and Confirm & issue plan	Print an error that the plan is stale; require staff to regenerate the plan; no units are issued from it.
EC14	Menu choice is non-numeric text (e.g., “abc”)	Print an error; show the menu again.
EC15	“View progress report” or “View expired/wasted report” chosen with no activity yet	Print the report with all counts and totals at zero.
EC16	Add Stock entry has an expiry date more than 35 days from today (e.g., 60 days out)	Print an error naming the date and the 35-day limit; unit not added — same treatment as a past-date rejection (EC6).
EC17	Add Stock entry has an expiry date exactly 35 days from today	Unit is accepted and added as a normal Available unit; the 35-day boundary itself is valid, not a rejection.
EC18	Add Stock entry has an expiry date exactly 36 days from today (one day past the cutoff)	Rejected the same as EC16 — confirms the boundary is strictly “35 or fewer days out,” not “around 35.” Unit not added.

Test Cases

Traces To	Input / Scenario	Expected Output
FR 3	Submit request with an invalid blood group. Blood group: "C+" (District Hospital, Govt, Qty 2, High)	Error: Invalid blood group "C+". Status: Rejected
FR 3	Submit request with a quantity that is zero, negative, or non-numeric. Quantity: "-3"	Error: Invalid quantity "-3". Quantity must be a positive number. Status: Rejected
FR 3	Submit request with an urgency outside High/Mid/Low. Urgency: "Urgent"	Error: Invalid urgency "Urgent". Must be High, Mid, or Low. Status: Rejected
FR 3	Submit request with a hospital type outside Private/Govt. Hospital type: "Clinic"	Error: Invalid hospital type "Clinic". Must be Private or Govt. Status: Rejected
FR 3	Submit request for a group with zero compatible stock available. Blood group: AB-, Qty 1, current AB- stock = 0, no compatible units	Requested: 1 units of AB- Available (compatible): 0 units Status: Rejected - insufficient stock (need 1, have 0) No units issued. No fee charged.
FR 2	Add stock with an expiry date already in the past. A+ 2025-01-01	Error: Expiry date 2025-01-01 is in the past. Unit not added.
FR 5	Choose "Confirm & issue plan" before any plan has been generated.	Error: No allocation plan has been generated yet. Choose option 3 (Generate allocation plan) first. No state change.
FR 4	Choose "Generate allocation plan" with zero pending requests.	No pending requests to process. Nothing generated.
FR 1 / FR 9	Choose a menu option outside 1-7 (e.g., 8, or blank input).	Error: Invalid option. Please choose a number between 1 and 7. (Stock table and menu are reprinted.)
FR 2	Add stock with a valid blood group and a future expiry date. O+ 2026-10-01	Added: U011 O+ (exp 2026-10-01) Unit appears as Available in the stock table.
FR 2	Add stock with a valid blood group and an expiry date more than 35 days from today. O+ 2026-12-15 (85 days from today)	Error: Expiry date 2026-12-15 is more than 35 days from today. Unit not added. Unit does not appear in the stock table.
FR 3	Submit a High urgency request with enough compatible stock; verify the bill is computed correctly. Blood group: O-, Qty 2, Govt, High	----- BILL ----- U0xx O- ... Rs 800 U0xx O- ... Rs 800 Transportation Charge: Rs 100 TOTAL AMOUNT: Rs 1700 Status: Fulfilled

Traces To	Input / Scenario	Expected Output
FR 4	Two pending requests: R1 (Low, submitted first), R2 (Mid, submitted second). Choose option 3.	Plan processes R2 (Mid) before R1 (Low), even though R1 was submitted first.
FR 5	A plan with one Fulfilled allocation (Total Rs 800) has been generated. Choose option 4.	Plan confirmed. Units issued, stock updated. Total revenue this session increases by Rs 800. Request removed from the pending queue.
FR 6	Unit U005 (A+, exp 2026-09-08) is still Available and its expiry date has now passed.	[1 unit expired and discarded: U005 (A+, exp 2026-09-08)] Unit no longer counted as Available in the stock table.
FR 7	3 Fulfilled and 2 Rejected requests have been handled so far. Choose option 5.	Requests handled: 5 total Fulfilled: 3 Rejected: 2 Total Revenue so far: Rs (sum of the 3 fulfilled bills)
FR 8	2 units have been wasted: U002 (A-, exp 2026-09-05), U008 (O+, exp 2026-09-06). Choose option 6.	Expired / Wasted Units: 2 U002 A- expired 2026-09-05 U008 O+ expired 2026-09-06 Total Value Lost: Rs 1000
FR 2	Add stock with an invalid group mixed with a valid one in the same entry. C+ 2026-10-01, O+ 2026-10-05	Error: Invalid blood group "C+". Unit not added. Added: U012 O+ (exp 2026-10-05)
FR 3	Submit a request where compatible stock is less than the quantity needed (partial shortage). Blood group: O+, Qty 5, available compatible = 3	Requested: 5 units of O+ Available (compatible): 3 units Status: Rejected - insufficient stock (need 5, have 3) No units issued. No fee charged.
FR 6	Unit U020 (A+) has an expiry date exactly equal to today's date.	Unit U020 stays status Available; it is not moved to Wasted; still counted in the stock table.
FR 4 / FR 5	A plan proposes unit U007 for request R1. Before "Confirm & issue plan" is chosen, U007 expires. Choose option 4.	Error: This plan is no longer valid (a proposed unit has expired). Regenerate the plan (option 3) before confirming. No units issued. No state change.
FR 1 / FR 9	Menu choice is non-numeric text. Choose an option: "abc"	Error: Invalid option. Please choose a number between 1 and 7. (Stock table and menu are reprinted.)
FR 7	Fresh session, no requests submitted yet. Choose option 5.	Requests handled: 0 total Fulfilled: 0 Rejected: 0 Total Revenue so far: Rs 0

Traces To	Input / Scenario	Expected Output
FR 8	Fresh session, no units wasted yet. Choose option 6.	Expired / Wasted Units: 0 Total Value Lost: Rs 0
FR 2	Add stock with an expiry date exactly 35 days from today (boundary case, should pass). O- (date typed = today + 35 days)	Added: U013 O- (exp = today + 35 days) Unit appears as Available — the 35-day boundary itself is accepted, not rejected.
FR 2	Add stock with an expiry date exactly 36 days from today (one day past the boundary, should fail). O- (date typed = today + 36 days)	Error: Expiry date is more than 35 days from today. Unit not added. Confirms the cutoff is strictly 35 days, not a rounded/approximate window.